

Marathon Project: Milestone 3 Business Report

Business Question #1: Merchandise Profitability per Race

Question: Which individual races generated the highest total profit from merchandise sales, and how does that profit compare to the number of items sold?

Query description:

1. Tables: RACES, MERCHANDISE, and RUNNER_PURCHASES.
2. Columns: RACE_NAME, QUANTITY, PRICE, and COST.
3. Joins: Joined RACES to MERCHANDISE on RACE_ID and MERCHANDISE to RUNNER_PURCHASES on PRODUCT_ID.
4. Logic: A column calculation $(m.PRICE - m.COST) * p.QUANTITY$ is used to determine the actual profit per transaction rather than just gross revenue. SUM and GROUP BY are used to aggregate these totals for each specific race.
5. Sorting: Results are sorted by TOTAL_PROFIT in descending order to immediately identify the most lucrative races.

SQL code:

```
SELECT
    r.RACE_NAME,
    SUM(p.QUANTITY) AS TOTAL_ITEMS_SOLD,
    SUM((m.PRICE - m.COST) * p.QUANTITY) AS TOTAL_PROFIT
FROM MSIS3333_MAIN.MARATHON_DB.RACES r
JOIN MSIS3333_MAIN.MARATHON_DB.MERCHANDISE m ON r.RACE_ID =
m.RACE_ID
JOIN MSIS3333_MAIN.MARATHON_DB.RUNNER_PURCHASES p ON
m.PRODUCT_ID = p.PRODUCT_ID
GROUP BY r.RACE_NAME
ORDER BY TOTAL_PROFIT DESC;
```

Results:

RACE_NAME	TOTAL_ITEMS_SOLD	TOTAL_PROFIT
Chi-town Dash Marathon	31868	\$569262.32
Surfs Up Marathon	17951	\$319037.49
Silicon Valley Sprint Half Marathon	13484	\$261556.16
Boston Tea Party Half Marathon	10908	\$197113.92
Philadelphia Half Marathon	10642	\$186111.58
Sprint the Sound Marathon	10143	\$180779.57
Indy Half Marathon	9212	\$165627.88
Research Triangle Half Marathon	8194	\$159201.06
Desert Skies Marathon	7617	\$149556.83
Philadelphia Marathon	8300	\$143249.00

Summary: These results suggest which specific race brands are most effective at converting runner participation into merchandise profit. This helps organizers decide which races deserve a larger inventory of premium items, and which may need a redesign of their merchandise strategy.

Business Question #2: Elite Runner Demographics by State

Question: Which states produce the highest number of "Elite" finishers (finish time under 180 minutes), and what is the average finish time for these runners per state?

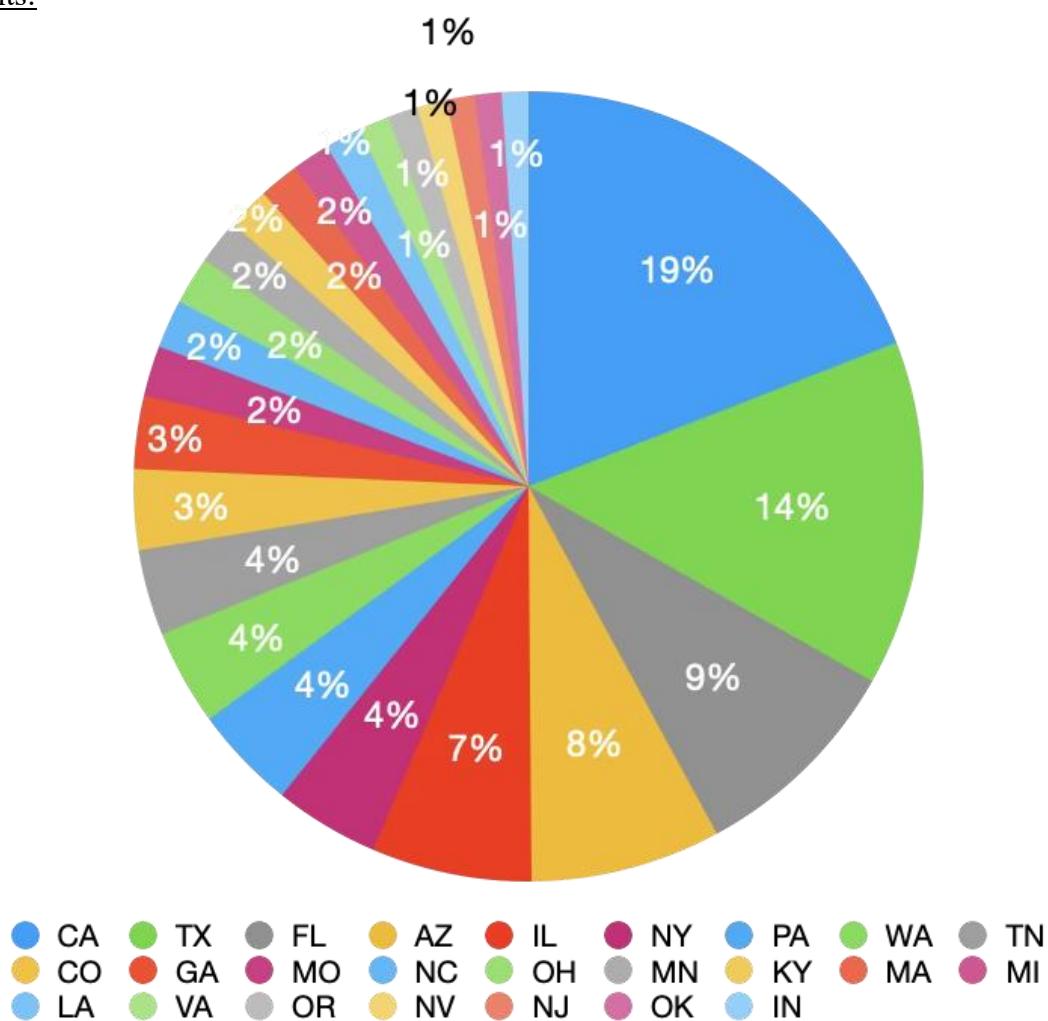
Query description:

1. Tables: RUNNERS, RACE_ROSTER, and RACES.
2. Columns: RESIDENCE_STATE, RUNNER_ID, FINISH_TIME, and DISTANCE.
3. Joins: Joined RUNNERS to RACE_ROSTER on RUNNER_ID and RACE_ROSTER to RACES on RACE_ID.
4. Logic: The WHERE clause filters for finish times under 180 minutes and ensures we are only looking at 'Marathon' distances to keep the comparison fair. We are specifically only looking at runners within the Nation.
5. Sorting: Results are limited to the top 25 States and then sorted by ELITE_RUNNER_COUNT descending to rank states by their "Elite" output.

SQL code:

```
SELECT TOP 25
    ru.RESIDENCE_STATE,
    COUNT(DISTINCT ro.RUNNER_ID) AS ELITE_RUNNER_COUNT,
    AVG(ro.FINISH_TIME) AS AVG_FINISH_TIME
FROM MSIS3333_MAIN.MARATHON_DB.RUNNERS ru
JOIN MSIS3333_MAIN.MARATHON_DB.RACE_ROSTER ro
    ON ru.RUNNER_ID = ro.RUNNER_ID
JOIN MSIS3333_MAIN.MARATHON_DB.RACES r
    ON ro.RACE_ID = r.RACE_ID
WHERE ro.FINISH_TIME < 180
    AND r.DISTANCE = 'Marathon'
    AND ru.RESIDENCE_COUNTRY != 'International'
GROUP BY ru.RESIDENCE_STATE
ORDER BY ELITE_RUNNER_COUNT DESC;
```

Results:



RESIDENCE_STATE	ELITE_RUNNER_COUNT	AVG_FINISH_TIME
CA	788	155.981645161
TX	584	155.652250379
FL	367	157.330237885
AZ	322	155.652173333
IL	271	154.832138264
NY	177	156.467278947
PA	176	156.372356481
WA	162	155.692494737
TN	145	158.128621302
CO	136	156.336094340
GA	124	155.160380952
MO	85	153.158483871
NC	82	154.928322222

OH	81	156.553136364
MN	77	156.544543478
KY	71	157.348051282
MA	68	158.306452055
MI	64	159.778380282
LA	61	159.422441176
VA	52	156.471228070
OR	51	155.568508772
NV	50	156.872561404
NJ	47	157.648870370
OK	46	156.297897959
IN	45	152.721872340

Summary: These results identify geographic "hotspots" for high-performance runners. Race organizers can use this data to target digital marketing spend or recruitment efforts in specific states that are likely to bring competitive prestige to the event.

Business Question #3: Event Financial Efficiency (Revenue vs. Expenses)

Question: For each race event, what was the Net Registration Income after subtracting all event expenses, and which events were the most financially efficient?

Query description:

1. Tables: RACE_EVENTS, RACES, RACE_ROSTER, and EXPENSES.
2. Columns: EVENT_NAME, REGISTRATION_FEE, and EXPENSE_AMOUNT.
3. Joins: Uses a Common Table Expression (CTE) to aggregate EXPENSES by EVENT_ID. The main query joins RACE_EVENTS to RACES and RACE_ROSTER to calculate gross registration revenue.
4. Logic: A CTE is used to prevent "double-counting" expenses when joining against the many rows of the RACE_ROSTER. The calculation $\text{SUM}(\text{REGISTRATION_FEE}) - \text{TOTAL_EXPENSES}$ provides the Net Income.
5. Sorting: Results are sorted by NET_INCOME descending to show which events are the most profitable after overhead.

SQL code:

```
WITH Event_Expenses AS (
    SELECT EVENT_ID, SUM(EXPENSE_AMOUNT) AS TOTAL_EXPENSES
    FROM MSIS3333_MAIN.MARATHON_DB.EXPENSES
    GROUP BY EVENT_ID
)
SELECT
    e.EVENT_NAME,
    SUM(r.REGISTRATION_FEE) AS GROSS_REG_REVENUE,
    ROUND(ee.TOTAL_EXPENSES,2) AS TOTAL_EXPENSES,
    ROUND((SUM(r.REGISTRATION_FEE) - ee.TOTAL_EXPENSES),2) AS
NET_INCOME
FROM MSIS3333_MAIN.MARATHON_DB.RACE_EVENTS e
JOIN MSIS3333_MAIN.MARATHON_DB.RACES r ON e.EVENT_ID =
r.RACE_EVENT_ID
JOIN MSIS3333_MAIN.MARATHON_DB.RACE_ROSTER ro ON r.RACE_ID =
ro.RACE_ID
JOIN Event_Expenses ee ON e.EVENT_ID = ee.EVENT_ID
GROUP BY e.EVENT_NAME, ee.TOTAL_EXPENSES
ORDER BY NET_INCOME DESC;
```

Results:

EVENT_NAME	GROSS_REG_REVENUE	TOTAL_EXPENSES	NET_INCOME
Chi-town Dash Race Weekend	1972165	1382841.69	589323.31
Surfs Up Race Weekend	1555884	972184.02	583699.98
Philadelphia Race Weekend	1435272	1034325.88	400946.12
Boston Tea Party Race Weekend	669630	456501.10	213128.90
Sprint the Sound Race Weekend	789558	622152.69	167405.31
Desert Skies Race Weekend	479950	339863.75	140086.25
Music City Race Weekend	739210	602561.59	136648.41
Research Triangle Race Weekend	598785	507901.33	90883.67
Miami International Race Weekend	622605	536964.05	85640.95
Silicon Valley Sprint Race Weekend	658764	637673.30	21090.70
Atlanta Premier Race Weekend	627589	609181.40	18407.60
Twin Cities Race Weekend	398432	389326.96	9105.04
Beach and Bay Race Weekend	449834	472569.81	-22735.81
Sweatin in Savannah Race Weekend	344513	388534.67	-44021.67
Gulf Coast Race Weekend	383658	429327.29	-45669.29
Indy Race Weekend	614376	669760.99	-55384.99
Bourbon Trail Race Weekend	513648	572935.98	-59287.98
Wild West Race Weekend	72432	139124.12	-66692.12
Hot Springs Trot Race Weekend	84560	154675.07	-70115.07
Steel City Race Weekend	476859	559546.63	-82687.63
Charleston Race Weekend	176519	293156.74	-116637.74
Big Easy Race Weekend	382115	508085.48	-125970.48
Kansas City Race Weekend	415119	548378.27	-133259.27
Big Sky Race Weekend	121744	283690.61	-161946.61
Flatiron Flyer Race Weekend	285504	472826.78	-187322.78
Santa Fe Sun Race Weekend	142752	403048.30	-260296.30

Summary: These results highlight the financial health of each event by weighing participant volume against operational costs. It allows organizers to see if certain high-revenue events are less efficient due to massive expenses, aiding in future budget allocations.